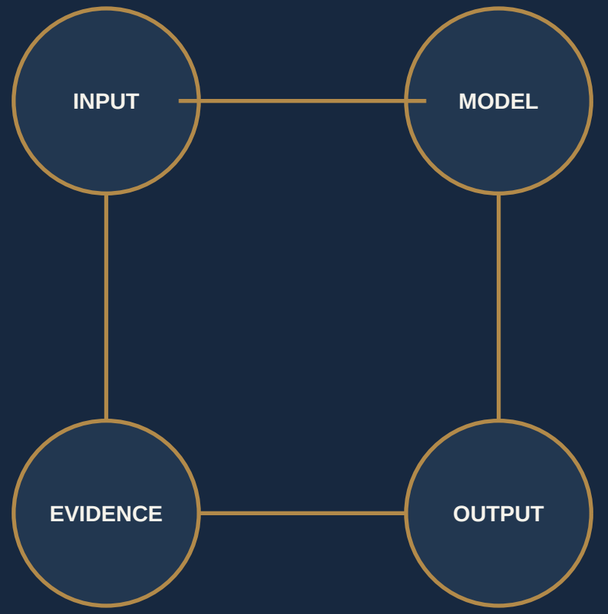
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

# Reinforcement Learning

AI.205 | University lecture series



# What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define agents.
- Apply key methods including states.
- Build or analyze a practical system involving actions.
- Evaluate assumptions, limitations, performance and responsible use.

# A connected learning journey

Each stage prepares the tools needed for the next.

01

## FOUNDATIONS

Agents

02

## METHODS

States

03

## SYSTEMS

Actions

04

## PRACTICE

Laboratory, evaluation and project

# The subject at a glance

A map of the ideas and techniques in this course

### 01 · Agents

Agents

### 02 · States

States

### 03 · Actions

Actions

### 04 · Rewards

Rewards

### 05 · Policies

Policies

### 06 · Value functions

Value functions

### 07 · Markov decision processes

Markov decision processes

### 08 · Q-learning

Q-learning

# Core ideas, made explicit

AI.205 • Distinguish definitions, assumptions and evidence

- Agents
- States
- Actions
- Rewards
- Policies

**DEFINE → MODEL → TEST → INTERPRET**

# Core ideas, made explicit

AI.205 • Distinguish definitions, assumptions and evidence

- Value functions
- Markov decision processes
- Q-learning
- Policy learning
- Exploration and exploitation

**DEFINE → MODEL → TEST → INTERPRET**

# Follow the system through its stages

Use the process diagram to connect components and outcomes.

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

# From concept to a real engineering decision

A short case study for discussion and worked-example narration

### SCENARIO

A real task involving agents.

### CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

### DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

# Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using agents.
- Implement or simulate a baseline using states.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

# Review · explain · apply

## KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

## PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

**Key takeaway: make the model visible, test it, and communicate its limits.**

Course-series slide design • detailed narration, demonstrations and assessments expand each module.